



Lecture 2

Web Programming

HTTP Protocol and HTML

The Backbone of Web Communication and Content Structure

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Overview of the HTTP Protocol

Definition: HTTP (HyperText Transfer Protocol) is the protocol used for transferring data over the web. It enables communication between web clients (browsers) and web servers.

Why It's Important: HTTP is the foundation of data exchange on the World Wide Web.

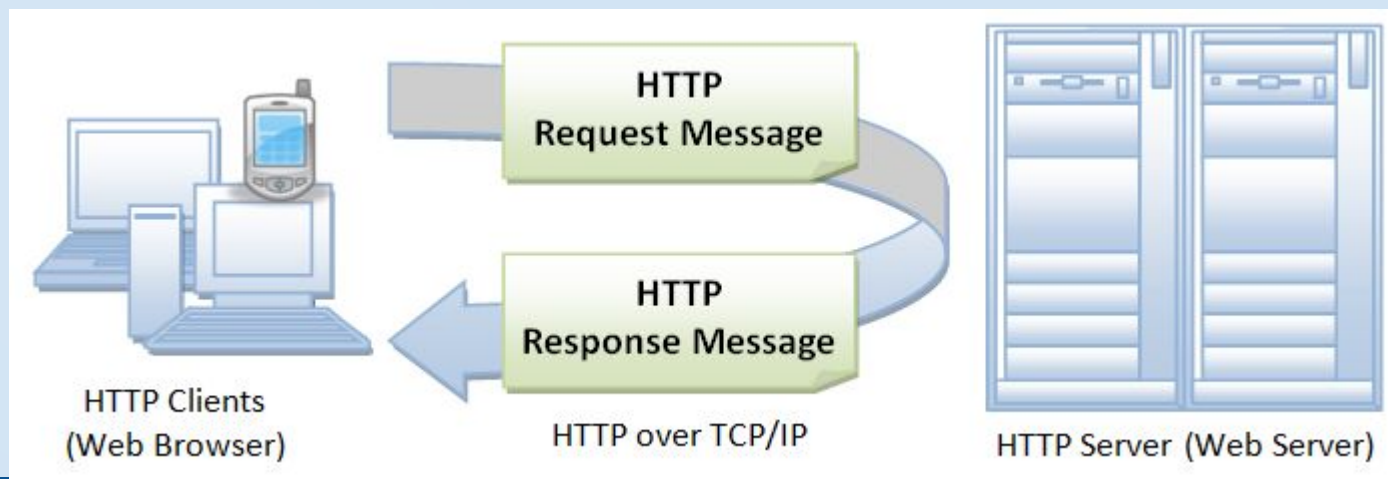
Key Point: HTTP is **stateless**, meaning each request is independent, and the server does not retain information about previous requests. Example.

How HTTP Works?

Request: A client (typically a web browser) sends an HTTP request to the web server. This request includes details like the type of request (GET, POST, etc.) and the resource (URL) being requested.

Processing: The server processes the request, retrieves the requested resource (HTML file, image, etc.), and prepares a response.

Response: The server sends an HTTP response back to the client, typically containing the requested resource or an error message if the resource cannot be found.





HTTP Request Structure

An HTTP request is made up of several parts:

- **Request Line:** Contains the HTTP method, the URL, and the HTTP version.
 - Example: `GET /index.html HTTP/1.1`
- **Headers:** Contain metadata about the request, such as the type of content requested or user-agent (browser).
 - Example: `User-Agent: Mozilla/5.0`
- **Body (optional):** Contains data sent with the request (e.g., form submissions).

Example:

```
GET /home HTTP/1.1
```

```
Host: www.example.com
```

```
User-Agent: Mozilla/5.0
```



HTTP Response Structure

The HTTP response sent from the server contains:

1. **Status Line:** Includes the HTTP version, status code, and a brief description of the status.
 - Example: `HTTP/1.1 200 OK`
2. **Headers:** Provide metadata about the response, such as content type, server information, or cookies.
 - Example: `Content-Type: text/html`
3. **Body:** The content of the response (HTML, images, JSON, etc.).

Example:

less

Copy code

```
HTTP/1.1 200 OK
```

```
Content-Type: text/html
```

```
Content-Length: 1234
```

```
<html>...</html>
```



HTTP Status Codes

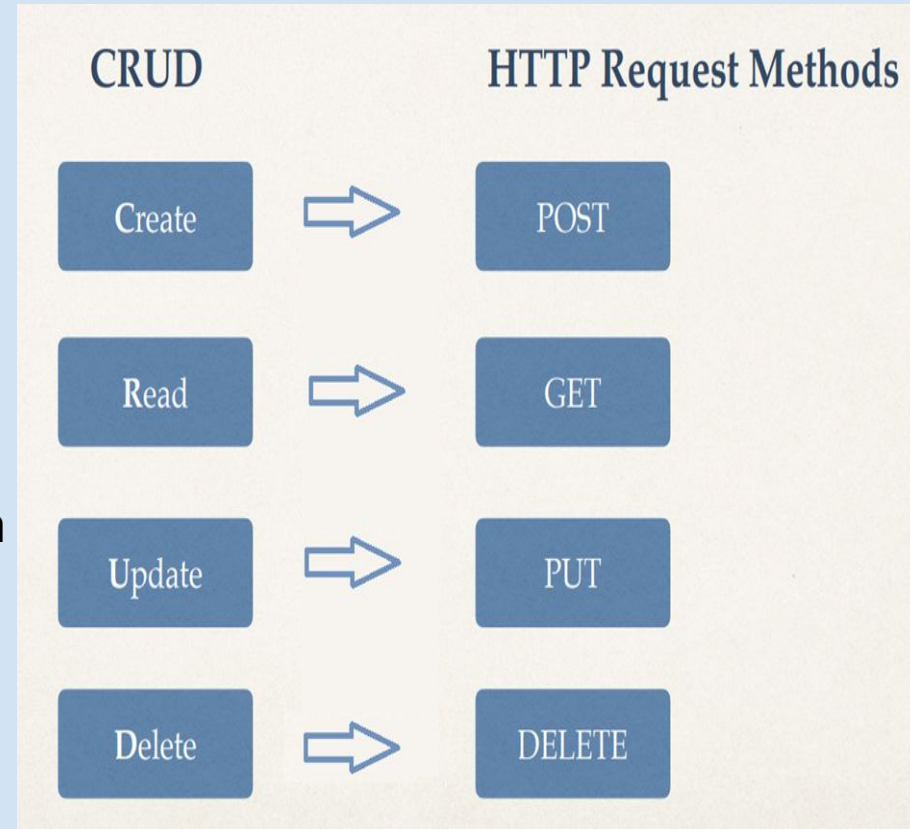
HTTP status codes indicate the result of a client's request. Some important status codes include:

- **200 OK:** The request was successful.
- **404 Not Found:** The requested resource could not be found on the server.
- **500 Internal Server Error:** There was an error on the server.
- **301 Moved Permanently:** The resource has been permanently moved to a new location.
- **403 Forbidden:** The server understands the request but refuses to authorize it.

HTTP Methods

HTTP defines several methods for interacting with resources:

1. **GET**: Retrieve a resource (e.g., an HTML file, image).
2. **POST**: Send data to the server (e.g., form submission).
3. **PUT**: Update a resource on the server.
4. **DELETE**: Remove a resource from the server.
5. **HEAD**: Retrieve only the headers (no body content).
6. **PATCH**: Apply partial modifications to a resource.





GET vs POST

- GET requests can be cached
- GET requests remain in the browser history
- GET requests can be bookmarked
- GET requests have length restrictions
- GET requests are to retrieve data
- POST requests are never cached
- POST requests do not remain in the browser history
- POST requests cannot be bookmarked
- POST requests have no restrictions on data length
- POST requests should be used to transmit data



Request Response Cycle

. GET <http://www.google.com> HTTP/1.1

Host: www.google.com

Request

. HTTP/1.1 200

Content-type: text/html

Content-length: xx

Response

<html>

<head> ... </head>

<body> Google </body>

</html>



Introduction to HTML

- **Definition:** HTML is the standard language used to create and design web pages.
- **Purpose:** It structures the content on a webpage, such as text, images, videos, and links.

Key Point: HTML uses **elements** (tags) to define the structure and content of a webpage.



HTML Elements Structures

Element comprises of:

- Start Tag
- End Tag
- Attributes and values
- Content

```
<tag attr1="value1" attr2="value2" ...> content...  
</tag>
```

```
<tag attr1="value1" attr2="value2" ... />
```



Basic HTML Structure

A basic HTML document includes several essential elements:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-scale=1.0">
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome to My Web Page</h1>
  <p>This is a paragraph of text.</p>
</body>
</html>
```

- **<!DOCTYPE html>:** Declares the document type.
- **<html>:** Root element of the document.
- **<head>:** Contains metadata about the webpage (title, character encoding).



Key HTML Elements

Text Elements:

- **<h1> to <h6>**: Headings (from largest to smallest).
- **<p>**: Paragraph.
- **<a>**: Anchor (for links).
- ****: Unordered list.
- ****: Ordered list.
- ****: List item.

Media Elements:

- ****: Embeds an image.
- **<audio>**: Embeds an audio file.
- **<video>**: Embeds a video file.

Form Elements:

- **<form>**: Defines a form for user input.
- **<input>**: Defines an input field.
- **<textarea>**: Defines a text area.



Heading and paragraph

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
..
<h6>This is heading 6</h6>
<p>My first paragraph.</p>
</body>
</html>
```



Links and navigation - Anchor Tag

```
<!DOCTYPE html>  
<html>  
<body>  
<h1>My First Heading</h1>  
<a href="https://www.wolframalpha.com/">This is a link</a>  
</body>  
</html>
```

Relative vs. Absolute URLs:

- **Relative URL:** Links to a resource within the same website (e.g., [page2.html](#)).
- **Absolute URL:** Links to a resource on any website (e.g., <https://www.example.com>).

NOTE: `target="_blank"` = > opens a new window



HTML Documents – image

```
<!DOCTYPE html>  
<html>  
<body>  
<h1>My First Heading</h1>  
  
</body>  
</html>
```

Note: src also have relative and absolute url



HTML Document - Buttons

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>HTML Buttons</h2>
```

```
<p>HTML buttons are defined with the button tag:</p>
```

```
<button>Click me</button>
```

```
</body>
```

```
</html>
```



HTML Forms

Forms allow users to submit data to a server:

- **<form>**: Defines the start of a form.
- **<input>**: Defines an input field (e.g., text box, button).
- **<label>**: Defines a label for input elements.
- **<select>**: Defines a drop-down menu.
- **<textarea>**: Defines a multi-line text input.

Example:

```
<form action="/submit" method="POST">  
  <label for="name">Name:</label>  
  <input type="text" id="name" name="name">  
  <input type="submit" value="Submit">  
</form>
```



Request method

URL for form submission

Text field

```
<html>
<body>
  <form method="GET" action="http://www.google.com/search">
    <input type="text" name="q" />
    <input type="submit" value="Google Search" />
  </form>
</body>
</html>
```

Represents a button that will submit form on click

Fa
Ahmed, F





Form parameters are shown in browser bar

The screenshot shows a Mozilla Firefox browser window. The title bar reads "world wide web - Google Search - Mozilla Firefox". The address bar shows the URL "https://www.google.com/search?q=world+wide+web". The search bar contains the text "world wide web". The search results page shows the top result from Wikipedia, titled "World Wide Web - Wikipedia, the free encyclopedia". The result snippet reads: "The **World Wide Web** (abbreviated as WWW or W3, commonly known as the web) is a system of interlinked hypertext documents accessed via the Internet." Below the snippet are links to "Internet" and "History of the World Wide Web".

Applications Places System

world wide web - Google Search - Mozilla Firefox

File Edit View History Bookmarks Tools Help

world wide web - Google Search

Most Visited (Untitled) Latest Headlines

Google world wide web

farooq.ahmad@nu.edu.pk

Web Images Books News Videos More Search tools

About 2,680,000,000 results (0.31 seconds)

[World Wide Web - Wikipedia, the free encyclopedia](#)
en.wikipedia.org/wiki/World_Wide_Web

The **World Wide Web** (abbreviated as WWW or W3, commonly known as the web) is a system of interlinked hypertext documents accessed via the Internet.

[Internet](#)
History of the Internet - World Wide Web - Leonard Kleinrock - TCP/IP

[History of the World Wide Web](#)
The World Wide Web ("WWW" or simply the "Web") is a global ...

[More results from wikipedia.org](#) »

[WorldWideWeb - Wikipedia, the free encyclopedia](#)
en.wikipedia.org/wiki/WorldWideWeb

WorldWideWeb (later renamed to Nexus to avoid confusion between the software and the **World Wide Web**) was the first web browser and editor. When it was ...

[News for world wide web](#)

[Robots test their own world wide web, dubbed RoboEarth](#)
BBC News - 6 days ago

Request method

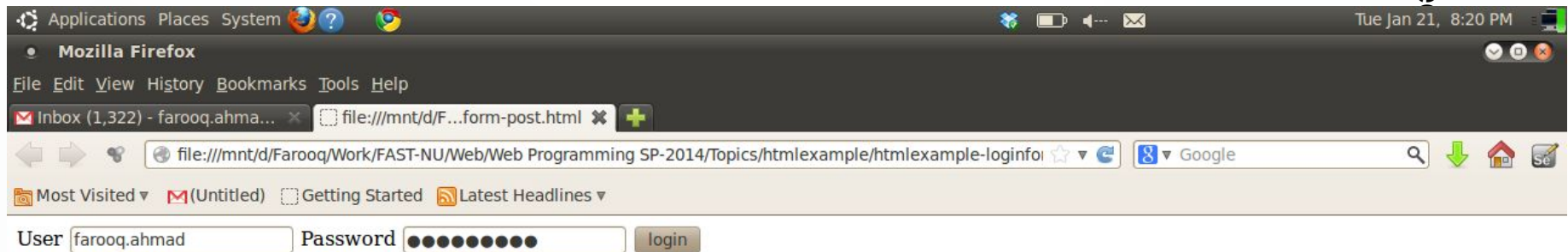
URL for form submission



Text field

```
<html>
<body>
  <form method="POST" action="http://slate.nu.edu.pk/portal/xlogin">
    User <input type="text" name="eid" />
    Password <input type="password" name="pw" />
    <input type="submit" value="login">
  </form>
</body>
</html>
```

Password field





Form parameters not shown

Applications Places System ? ? ? Tue Jan 21, 8:23 PM

SLATE -- Sakai Learning and Teaching Enviroment : My Workspace : Home - Mozilla Firefox

File Edit View History Bookmarks Tools Help

Inbox (1,322) - farooq.ahma... SLATE -- Sakai Learning and ...

slate.nu.edu.pk/portal Google

Most Visited (Untitled) Getting Started Latest Headlines

slate
Sakai learning and teaching environment

Logout

My Workspace Object Oriented Analysis & Design CS 2013-03 LHR Web Programming CS 2013-03 LHR Object Oriented Analysis & Design CS 2014-01 LHR

Message Of The Day Options

Home Profile Membership Schedule Resources Announcements Workspace Setup Preferences Account Help

Spring 2014 courses are being uploaded for all campuses of the university, if your registered courses are not showing up on SLATE, then make sure that they are updated on NeON

SLATE (slate.nu.edu.pk) is fully functional for all campuses of the university.

[Video Tutorial] [Why my course tab is not appearing on SLATE?](#)

Customize (show/hide, change order etc) previous semester course tabs by following steps or [watch video tutorial](#)

1. Log into **SLATE**
2. **My Workspace** -> **Preferences** -> **Customize Tabs**
3. In the **My Active Sites** box, **select a site** you want to move.
4. Use the arrow buttons to move the order of the site.
5. Repeat for any additional sites.
6. You can also increase the number of tabs SLATE will display or even hide old or unnecessary sites.
7. Click **Update Preferences** to save your changes and to view the change

Calendar Options

January, 2014

Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1

Message Center Notifications

Recent Announcements

Waiting for slate.nu.edu.pk...

[Zotero] [Sema...] [jpeg2...] [Devel...] [htmlx...] [Lesson...] [HTML...] [SLATE ...] [OAO ...] [03 Obj...] [web ar...]



Types of Input

- **1. Text**

- Allows the user to enter a single line of plain text.

- `<input type="text" placeholder="Enter name">`

- **2. Password**

- Hides the text.

- `<input type="password" placeholder="Password">`

- **3. File**

- Let's the user upload a file. A Filepicker is displayed.

- `<input type="file" accept="image/*">`



Types of Input

- **1. Checkbox**

- Represents a binary choice (checked/unchecked).

- `<input type="checkbox" id="subscribe">`

- **2. Radio**

- Allows selecting one option from a group of options. Multiple radio buttons that form part of the same group will have the same name value.

- `<input type="radio" name="gender" value="male">`
`<input type="radio" name="gender" value="female">`

- **3. Date**

- Allows a user to select a date using a datepicker.

- `<input type="date">`.



HTML Tables

HTML tables are used to display data in a grid format.

- **<table>**: Defines the table.
- **<tr>**: Table row.
- **<td>**: Table data (cell).
- **<th>**: Table header.

```
<table>
```

```
  <tr>
```

```
    <th>Name</th>
```

```
    <th>Age</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>John</td>
```

```
    <td>30</td>
```

```
  </tr>
```

```
</table>
```



Table

To make a cell span over multiple rows, use the **rowspan** attribute

To make a cell span over multiple columns, use the **colspan** attribute:

NAME		

APRIL		

```
<table>
  <tr>
    <td rowspan=3>April</td>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
    <td></td>
  </tr>
</table>
```



Block Level Elements

These elements take up the full width of their parent container and start on a new line. They stack vertically and usually occupy the entire available horizontal space.

Examples:

`<div>`

`<p>`

`<h1>`, `<h2>`, `<h3>`, etc.



Inline Elements

These elements only take up as much width as their content. They do not start on a new line and only occupy the space they need, staying within the flow of the content around them.

Examples:

`<span>`

`<label>`

`<img>`



HTML Comment Tags

```
<!DOCTYPE html>
```

```
<html>
```

```
<body style="background-color : powderblue;">
```

```
<!-- Write your comments here -->
```

```
<h1 style="text-align:center;">Centered Heading</h1>
```

```
<p style="text-align:center;">Centered paragraph.</p>
```

```
</body>
```

```
</html>
```

HTML Document - list

► Ordered List

`<ol>`

`<li>Coffee</li>`

`<li>Tea</li>`

`<li>Milk</li>`

`</ol>`

► Unordered List

`<ul>`

`<li>Coffee</li>`

`<li>Tea</li>`

`<li>Milk</li>`

`</ul>`



HTML5 Features

HTML5 introduced new features for better web experiences:

1. **<article>**: Represents a self-contained piece of content.
2. **<section>**: Represents a section of a document.
3. **<nav>**: Defines navigation links.
4. **<video>** and **<audio>**: Embed media content without needing plugins.



Enclosing Tags

These tags require both opening and closing tags.

They are used for elements that enclose content and can contain child elements.

Format: <tagName>content</tagName>

```
<p>This is a paragraph.</p>
<div>
  <h1>Heading</h1>
  <p>Some text inside a div.</p>
</div>
<button>Click Me</button>
```



Self-closing Tags

These tags do not need a separate closing tag.

They are typically used for elements that cannot contain child elements.

Format: <tagName />

```
  
<input type="text" />  
<br />
```



Class Test

- ▶ Create an unordered list containing the names of any three provinces of Pakistan. Then, create three nested ordered lists within the unordered list, containing names of any two cities of each province.